Incident Escalation Practice Documentation

The objective of this task is to simulate a Security Operations Center (SOC) incident escalation process. The activity includes creating an incident case, escalating it to Tier-2 analysts, drafting a Situation Report (SITREP), and demonstrating an automated workflow for handling high-priority alerts.

2. Tools Used

- TheHive
- Splunk Phantom
- Google Docs

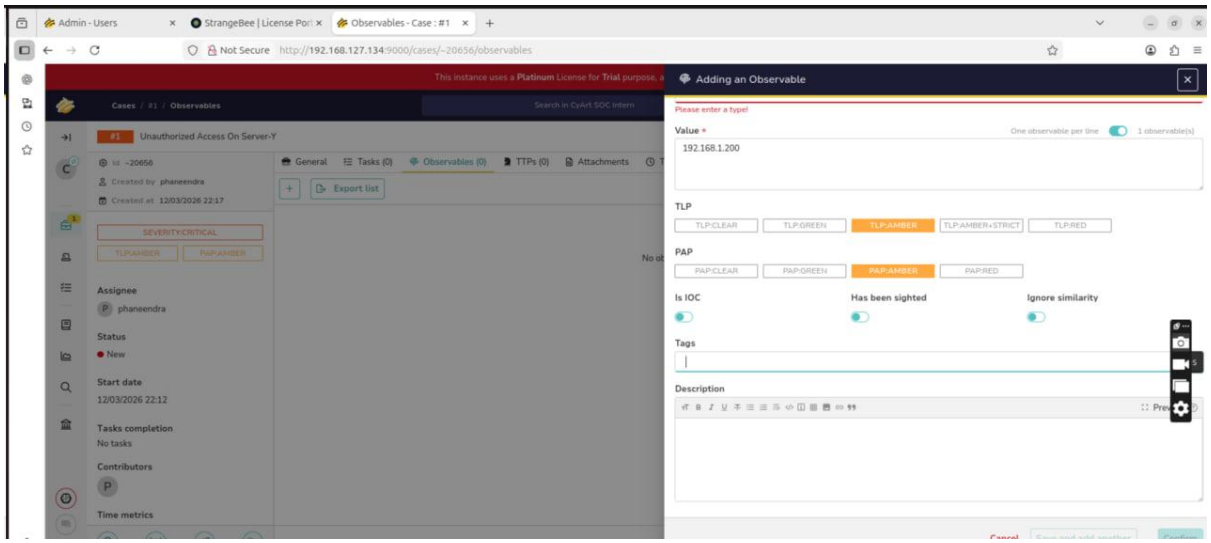
3. Escalation Simulation

- Step 1: Access TheHive
- Open the TheHive dashboard in the browser.
- Example:
- http://localhost:9000 ⍰
- Login using administrator credentials.
- Step 2: Create Incident Case
- Navigate to Cases → Create New Case.
- Case details used in the simulation:

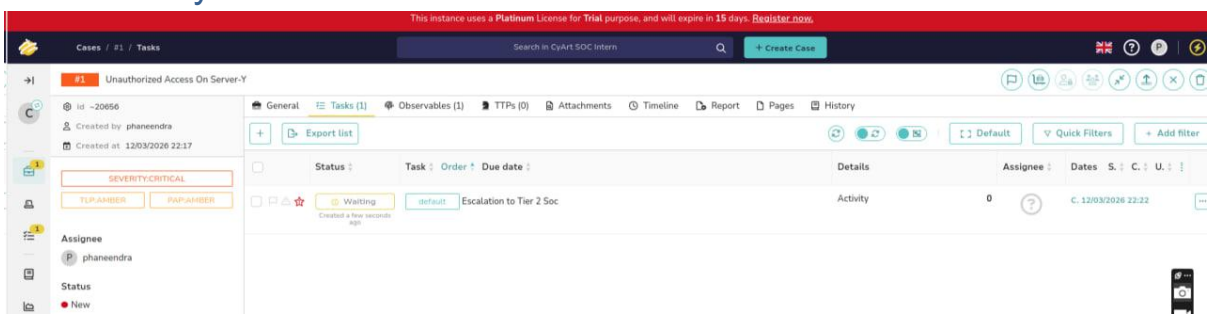
Title
Unauthorized Access on Server-Y
Severity
High

Description
Multiple SSH authentication failures were detected targeting Server-Y from IP address 192.168.1.200. The activity indicates a potential unauthorized access attempt.
MITRE ATT&CK Technique
T1078 – Valid Accounts
Step 3: Add Observable

Observable Type
IP Address
Observable Value
192.168.1.200
Tag
Suspicious activity
This observable helps analysts track the source of the attack.



Step 4: Escalate Incident
A task was created within the case to escalate the investigation.
Task Name
Escalation to Tier-2 SOC Team
Description
The alert indicates repeated authentication failures from an external IP address. Further investigation is required by the Tier-2 security team to analyze logs and determine if credential abuse or brute-force activity occurred.



4. Escalation Summary (100 Words)

The SOC monitoring system detected multiple unauthorized SSH login attempts targeting Server-Y from IP address 192.168.1.200 at 13:00 on 18 August 2025. The activity indicates a potential credential abuse attempt mapped to MITRE ATT&CK technique T1078 (Valid Accounts). Initial investigation confirmed repeated authentication failures within a short time window. To prevent possible compromise, the affected server was temporarily isolated from the network. Security logs were analyzed to verify the suspicious activity. The incident was escalated to the Tier-2 SOC team for deeper analysis and investigation, including reviewing authentication logs and identifying possible lateral movement attempts within the network.

4. Situation Report (SITREP)

Title - Unauthorized Access on Server-Y

An unauthorized access attempt was detected on Server-Y at 13:00 on 18 August 2025. The attack originated from IP address 192.168.1.200 and involved repeated SSH authentication failures. The activity corresponds to MITRE ATT&CK technique T1078, which indicates potential misuse of valid credentials.

Actions Taken

The affected server was temporarily isolated to prevent further unauthorized access. Security logs were reviewed to confirm the suspicious activity. The incident was escalated to the Tier-2 SOC team for deeper investigation and further response actions.

6. Workflow Automation

Automation was simulated using Splunk Phantom playbooks.

Alert Received
↓
Check Alert Severity
↓
If Severity = High

↓
Automatically Assign Case to Tier-2 Analyst
↓
Notify SOC Team

This automation reduces response time and ensures high-priority alerts are immediately handled by experienced analysts.

7. Mock Alert Test

A test alert was created to simulate the automation process.

Alert Name - Unauthorized SSH Access
Source IP - 192.168.1.200
Severity -High
Result - The system automatically assigned the alert to the Tier-2 SOC team and created an incident case for investigation.